When descriptive forecast-horizon gains do not imply global evidence of incremental meteorological predictability: a multi-site PM_{10} study

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Abstract

Forecast skill relative to persistence is often summarised by how long an improvement lasts, but a longer skill horizon does not necessarily provide equally strong evidence that an added information set contributes incremental predictive information. We evaluate hourly $\mathbf{PM_{10}}$ forecasts at nine sites (Madrid Casa de Campo and eight Irish stations) over January–July 2023 using an expanding-window rolling-origin scheme with 24-h spacing between forecast origins and horizons $\boldsymbol{h = 1, \ldots, 24}$. The incremental comparison uses the same direct multi-horizon XGBoost learner with two information sets: pollutant lags alone and pollutant lags plus meteorological observations available at the forecast origin. This is an observational information-set experiment, not an evaluation using future numerical weather prediction (NWP) forecasts. In Madrid, adding meteorology extends the strict longest-contiguous positive persistence-relative skill horizon from 9 to 17 h (+8 h), whereas the relaxed last-positive-skill horizon increases from 15 to 17 h (+2 h). However, none of the 36 dimensionally corrected comparisons using the Diebold–Mariano test with Harvey–Leybourne–Newbold correction (DM-HLN) at $\boldsymbol{h \in \{1, 6, 12, 24\}}$ survives global Bonferroni correction under either the overlap-based specification ($\boldsymbol{q_{\text{overlap}} = 0}$; 0/36) or a fixed automatic Newey–West sensitivity allowing positive-lag dependence (0/36). Thus, descriptive horizon extension is evident at some sites, but family-wise global support for attributing those gains to incremental meteorological information remains limited.

Keywords: PM10, rolling-origin evaluation, persistence-relative skill, forecast horizon, Diebold–Mariano test, multiplicity

1 Introduction

The usefulness of an air-quality forecast depends strongly on lead time. A model may improve near-term accuracy while offering little advantage later, so an hourly PM_{10} forecast is useful only for as long as it remains better than a simple persistence reference. Persistence is therefore not merely a convenient baseline. It provides a demanding test of whether a forecasting system adds information beyond the most recent observation in a setting where concentrations reflect emissions together with meteorologically driven transport and mixing (Seinfeld and Pandis 2006; Kumar et al. 2016).

That comparison becomes less straightforward when meteorological covariates are added. Pollutant history already contains information about some of the same underlying processes, so a reduction in forecast error after adding meteorology may reflect genuinely incremental information, a different representation of structure already present in lagged PM_{10}, or a mixture of both. The balance can also change with site and lead time. Published forecasting gains therefore span heterogeneous designs and outcomes (Méndez et al. 2023; Pak et al. 2018; Bai et al. 2018; Ansari and Quaff 2025).

The central distinction is between a descriptive and an inferential claim. A model may remain better than persistence over more horizons after meteorology is added, making the extension descriptively meaningful. That observation alone, however, does not establish how strongly the added information is supported once sampling uncertainty and multiple comparisons are considered.

We examine this distinction in a nine-site hourly PM_{10} experiment comprising Madrid Casa de Campo and eight Irish stations, evaluated over January–July 2023 with an expanding-window rolling-origin design. Persistence and SARIMA serve as reference benchmarks. The incremental comparison holds the learner fixed and contrasts direct multi-horizon XGBoost based on pollutant lags with the same model augmented by meteorological observations available at the forecast origin. We summarise horizon-wise performance using persistence-relative RMSE skill curves and the descriptive horizon measures H^*, and assess inferential attribution with DM-HLN tests (Diebold and Mariano 1995; Harvey et al. 1997) at $h \in \{1, 6, 12, 24\}$ under global multiplicity control across the 36 planned site–horizon comparisons.

This design reveals an important separation between the two levels of evidence. Some sites show sizeable descriptive extensions of persistence-relative skill, yet these gains provide weak family-wise global support for incremental meteorological predictability after accounting for the planned inferential family. Allowing residual positive-lag dependence through a fixed automatic Newey–West sensitivity leaves that global conclusion unchanged, although the limited local signals depend more on the specification.

2 Data

2.1 Madrid

Hourly PM_{10} concentrations and concurrent meteorological measurements were obtained for Madrid Casa de Campo from the Madrid City Council open-data portal. The station is classified as an urban-background site on the western edge of the city, away from major point sources, and is representative of citywide background conditions. The study period spans January 2020 through July 2023; records with missing PM_{10} or meteorological data were flagged and excluded from model training and evaluation on a per-origin basis. Seven meteorological variables were retained: precipitation (*rain*), temperature (*temp*), relative humidity (*rhum*), mean sea-level pressure (*msl*), wind speed (*wdsp*), wind direction (*wddir*), and solar radiation.

2.2 Ireland

Hourly PM_{10} concentrations and meteorological covariates were obtained from the Irish Environmental Protection Agency (EPA) monitoring network. Eight stations distributed across Ireland were selected on the basis of data completeness over the study window (January 2020 – July 2023): Birr (Co. Offaly), Dublin Airport, Dundalk (Co. Louth), Pearse Street Dublin, Ringsend Dublin, Edenderry (Co. Offaly), Henry Street Limerick, and Portlaoise (Co. Laois). Meteorological variables were extracted from the nearest mapped EPA synoptic station for each monitoring site.

Table 1 summarises the basic PM_{10} statistics for all nine sites over the training and evaluation periods. Concentrations are generally low to moderate, with training-period means ranging from 11.3 $\mu g\,m^{-3}$ (Portlaoise) to 19.1 $\mu g\,m^{-3}$ (Dublin Airport), and Madrid at 18.4 $\mu g\,m^{-3}$. The training-period 95th percentile exceeds 40 $\mu g\,m^{-3}$ at four of nine sites: Madrid, Dublin Airport, Ringsend Dublin, and Edenderry. Data completeness is high: missing rates are below 2% at all sites during both periods.

Table 1 Descriptive statistics of hourly PM_{10} ($\mu g\,m^{-3}$) for all nine sites. Training: 2020–2022; evaluation: Jan–Jul 2023. n: valid hourly observations; miss: missing rate (%).

	Training (2020–2022)				Evaluation (2023)			
Station	n	Mean	SD	P95	n	Mean	SD	P95
Madrid	25 630	18.4	20.0	42.0	4 988	15.9	12.3	34.6
Birr	20 372	12.8	12.5	31.4	5 085	13.2	11.3	31.2
Dublin Airport	21 958	19.1	16.1	46.9	3 504	15.7	10.9	34.3
Dundalk	21 938	11.4	8.6	24.3	3 658	13.1	10.2	30.7
Pearse St. Dublin	17 009	13.0	9.9	30.2	5 088	15.4	10.1	33.3
Ringsend Dublin	21 949	15.6	16.1	41.9	5 087	14.0	13.1	36.7
Edenderry	11 526	17.8	19.8	48.8	5 029	15.9	16.0	38.5
Henry St. Limerick	13 128	13.0	12.1	30.8	5 052	11.9	8.6	27.2
Portlaoise	21 958	11.3	9.7	27.5	5 088	10.9	7.9	25.1

3 Methods

3.1 Backtesting protocol

We adopt a rolling-origin expanding-window backtesting scheme to preserve prospective temporal information flow and strict separation between training and evaluation data. For each forecast origin t, the training window covers all available observations up to (but not including) t; no future information is used at any point. The forecast origin advances in daily strides of 24 h, producing one $h = 1, \dots, 24$ forecast vector per day.

The evaluation period was 1 January–31 July 2023, with forecast origins spaced by 24 h. The number of valid evaluated origins varied by site according to data availability. The mandatory minimum training window is 8 760 observations (one year), so origins in early 2023 draw on three years of history. This protocol prevents look-ahead from future observations in the rolling-origin evaluation. Model configurations and feature sets used in the evaluation were held fixed across evaluation origins; no origin-level tuning or cross-validation was performed.

3.2 Model specifications

3.2.1 Persistence.

The persistence model predicts $\hat{y}_{t+h} = y_t$ for all h. It serves as the denominator benchmark for the skill score.

SARIMA.

We fit a SARIMA$(1,0,1)(1,0,0)_{24}$ model at each origin using the *statsmodels* implementation of exact maximum likelihood. To avoid numerical instability of the Kalman filter on very long training windows, the training series is truncated to the most recent 17 520 observations (approximately two calendar years) at each origin. Multi-step-ahead forecasts are generated by iterating the fitted state-space model forward. SARIMA is treated as a univariate statistical reference; it does not receive meteorological inputs.

XGBoost — lags only.

A separate XGBoost regressor (Chen and Guestrin 2016) is trained for each horizon $h \in \{1, \dots, 24\}$ (the *direct* multi-horizon strategy). The feature set consists of eight lagged PM_{10} values $(y_{t-1}, y_{t-2}, y_{t-3}, y_{t-6}, y_{t-12}, y_{t-24}, y_{t-48}, y_{t-168})$ and four calendar indicators (hour of day, day of week, month, Julian day). Hyperparameters are fixed across all origins and horizons (Table 2) to ensure fair comparison; no origin-level cross-validation is performed.

XGBoost — lags + meteorology.

Identical to the lags-only model except that the meteorological variables measured at time t are appended to the feature vector (seven for Madrid, nine for the Irish stations). This is an *observational* setup in which meteorological information is available at the moment of forecast

Table 2 Fixed XGBoost hyperparameters used for all models and horizons.

Parameter	Value
Number of estimators	300
Maximum tree depth	4
Learning rate	0.05
Row subsampling	0.90
Column subsampling	0.90
Objective	`reg:squarederror`
Random seed	42

issuance. Madrid uses concurrent station observations, whereas the Irish covariates come from the nearest mapped EPA synoptic station. Because observed meteorological values are used rather than numerical weather prediction (NWP) forecasts, the reported skill scores should not be interpreted as operational performance with NWP inputs (see Section 5).

3.3 Evaluation metrics

3.3.1 Skill score.

For each model m and horizon h, the RMSE skill score relative to persistence is:

$$S_m(h) = 1 - \frac{\mathrm{RMSE}_m(h)}{\mathrm{RMSE}_{\mathrm{pers}}(h)}, \tag{1}$$

where the RMSEs are computed over all forecast origins in the evaluation period. Positive $S_m(h)$ indicates that model m outperforms persistence at horizon h.

Useful forecast horizon H^*.

We report two descriptive persistence-relative horizon summaries based on the sign of $S_m(h)$ over $h \in \{1, \ldots, 24\}$:

- $H^*_{\textbf{strict}}$ **(primary)**: the length of the longest *contiguous* run of horizons with $S_m(h) > 0$.
- $H^*_{\textbf{relax}}$ **(secondary)**: the last horizon h for which $S_m(h) > 0$.

H^* is a descriptor of the skill-curve sign pattern and is not itself a hypothesis test.

Diebold–Mariano test (DM-HLN), dependence specification, and multiplicity.

Incremental meteorological information is assessed inferentially via horizon-wise Diebold–Mariano tests (Diebold and Mariano 1995) with the Harvey–Leybourne–Newbold finite-sample correction (Harvey et al. 1997) (DM-HLN), comparing squared-error losses of XGBoost (lags only) versus XGBoost (lags + meteorology) at physical horizons $h \in \{1, 6, 12, 24\}$.

Because forecasts are issued once per day (24-h origin spacing), the loss-differential sequence advances in daily steps rather than in physical hours. We therefore define $h_{\mathrm{DM}} = \lceil h/24 \rceil$ as the horizon expressed in loss-differential steps. For $h \in \{1, 6, 12, 24\}$, $h_{\mathrm{DM}} = 1$ for all tests, so the mechanically induced overlap bandwidth is $q_{\mathrm{overlap}} = \max(0, h_{\mathrm{DM}} - 1) = 0$. This overlap-based choice removes the *mechanically required* positive-lag covariance from overlapping multi-step forecast errors, but it does not imply that loss differentials are independent or that residual dependence cannot exist.

To address potential positive-lag dependence not induced mechanically by overlap, we also report a fixed automatic Newey–West sensitivity analysis (Newey and West 1987) using the bandwidth rule $q_{\mathrm{auto}} = \lfloor 4\,(n/100)^{2/9} \rfloor$.

The primary inferential family comprises the 36 planned site–horizon comparisons (9 sites × 4 horizons), controlled with global Bonferroni correction. As a within-station sensitivity analysis addressing a narrower inferential scope, we also apply Benjamini–Hochberg false-discovery-rate control separately within each station (4 horizons per station).

4 Results

4.1 Madrid—substantial descriptive horizon extension

Madrid provides the clearest descriptive contrast between the two XGBoost information sets. The lags-only model remains better than persistence over only part of the 24-h range, whereas adding meteorological observations extends positive persistence-relative skill further into the day (Figure 1). The primary strict descriptor captures the largest change: the longest contiguous positive-skill run increases from $H^*_{\mathrm{strict}} = 9$ h with lags only to $H^*_{\mathrm{strict}} = 17$ h with lags + meteorology, an extension of +8 h. The relaxed descriptor changes less, from 15 to 17 h (+2 h). These summaries answer different descriptive questions. The strict measure reflects continuity of positive skill, whereas the relaxed measure records the last horizon for which $S(h) > 0$. Reporting both therefore shows how the apparent size of the horizon gain depends on the feature of the skill profile being summarised.

4.2 Cross-site heterogeneity and limited observable headroom

The Irish stations show a different pattern. Under the lags-only specification, persistence-relative skill remains positive across much of the evaluated range at several sites, leaving relatively little room for the strict horizon to increase when meteorology is added (Figure 2). The mean strict horizon increases from 21.875 to 22.875 h, a mean gain of +1.000 h, while five of the eight stations already attain $H^*_{\mathrm{strict}} = 24$ h with lags only. Across both information sets, 11 station/condition cases reach $H^*_{\mathrm{strict}} = 24$ h. These values reach the design boundary and should therefore be treated as right-censored by $H_{\max} = 24$, not as estimates of a natural predictability limit.

Table 3 gives the corresponding station-level horizon summaries. The contrast between Madrid and the Irish stations also shows why an "hour gained" is not directly comparable across sites: the amount of observable headroom depends on the underlying lags-only skill profile. This makes inferential evidence for incremental meteorological information a necessary complement to the descriptive horizon summaries.

4.3 Descriptive gains do not scale to family-wise global support

The paired loss-differential tests give a different view of the same comparison. Under the primary dimensionally coherent overlap-based DM-HLN specification ($q_{\mathrm{overlap}} = 0$), all 36 planned station–horizon tests are valid (Table 4). Three comparisons have raw $p < 0.05$, but none survives global Bonferroni correction across the 36-test family (0/36). At the narrower within-station level, Benjamini–Hochberg control retains three signals: Birr at $h = 1$, Madrid at $h = 12$, and Pearse St. Dublin at $h = 6$. There is no contradiction between these findings. Local signals can remain after within-station correction even when the planned family-wise analysis yields no rejection, and the global 0/36 result should not be read as evidence of equivalence or proof of no effect.

4.4 Global conclusion remains unchanged under fixed automatic Newey–West sensitivity

The overlap-based specification $q_{\mathrm{overlap}} = 0$ accounts for dependence mechanically induced by forecast overlap under 24-h origin spacing, but it does not rule out residual positive-lag dependence in the loss differentials. We therefore repeat the analysis using the fixed automatic Newey–West sensitivity reported in Table 5. The family-wise result does not change: no comparison survives global Bonferroni correction (0/36). Three within-station Benjamini–Hochberg signals again remain, although the set is not identical. Birr at $h = 1$ and Pearse St. Dublin at $h = 6$ are retained under both specifications. Madrid at $h = 12$ appears only with the overlap-based specification, whereas Pearse St. Dublin at $h = 12$ appears only with the automatic Newey–West sensitivity.

The global conclusion is therefore stable across the two audited dependence specifications, while the small set of local signals is more sensitive to that choice.

The complete cell-level inferential landscape is shown in Figure 3; the corresponding numerical ledger is provided in Supplementary Table S1.

5 Discussion

Madrid illustrates why descriptive skill horizons are informative but not sufficient for attribution. Adding meteorological observations extends the strict positive-skill run from 9 to 17 h, a substantial change relative to persistence. The relaxed measure, however, moves only from 15 to 17 h. Even within one site, the apparent size of a "horizon extension" therefore depends on whether the quantity of interest is continuity of positive skill or the last horizon at which positive skill occurs.

Ireland exposes a different limitation of the same summary. Several lags-only skill curves

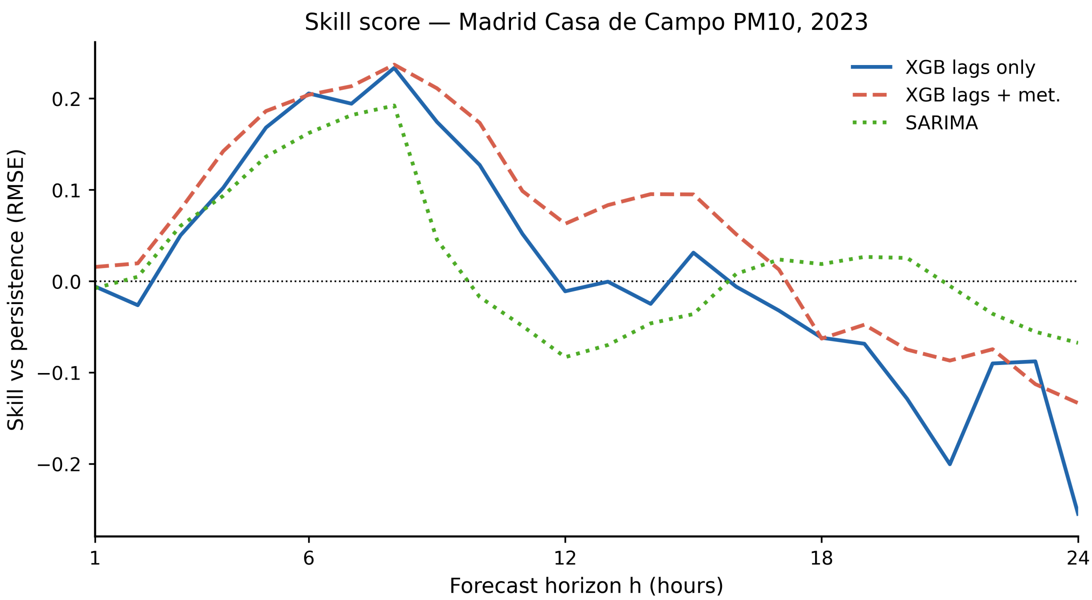

Fig. 1 Madrid Casa de Campo (Jan–Jul 2023): persistence-relative RMSE skill $S_m(h) = 1 - \mathrm{RMSE}_m(h)/\mathrm{RMSE}_{\mathrm{pers}}(h)$ over physical horizons $h = 1, \ldots, 24$ h. Positive values indicate lower RMSE than persistence. Blue solid: XGBoost (lags only). Red dashed: XGBoost (lags + meteorology, using meteorological observations available at the forecast origin). Green dotted: SARIMA(1,0,1)(1,0,0)$_{24}$. Horizontal line: $S = 0$.

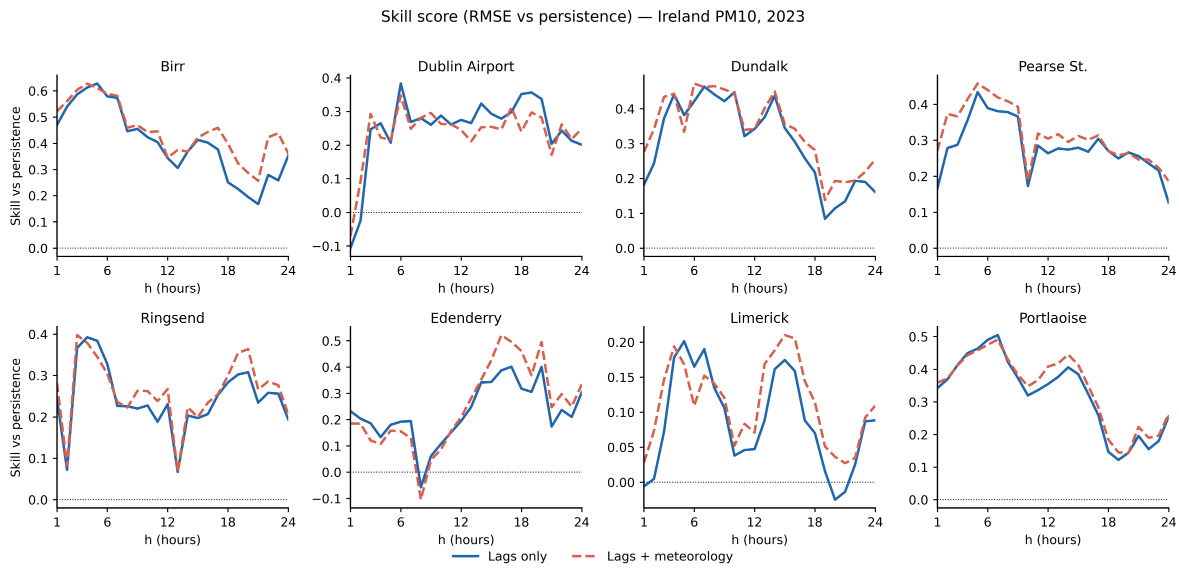

Fig. 2 Ireland (Jan–Jul 2023): persistence-relative RMSE skill $S_m(h) = 1 - \mathrm{RMSE}_m(h)/\mathrm{RMSE}_{\mathrm{pers}}(h)$ over physical horizons $h = 1, \ldots, 24$ h for eight stations (one panel per station). Blue solid: XGBoost (lags only). Red dashed: XGBoost (lags + meteorology, using meteorological observations available at the forecast origin). Horizontal line: $S = 0$. Stations with $H^*_{\mathrm{strict}} = 24$ reach the 24-h evaluation boundary.

remain positive through most of the 24-h evaluation range, so the strict descriptor often reaches $H_{\max} = 24$ before meteorology is added. The mean strict horizon increases only from 21.875

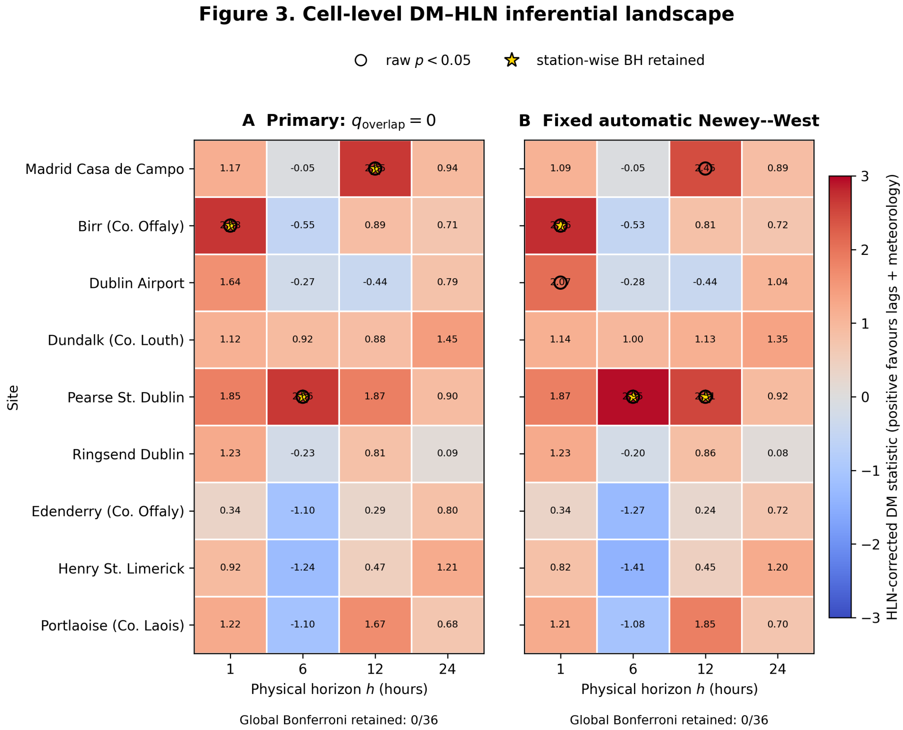

Fig. 3 Cell-level DM–HLN inferential landscape for the 36 planned site–horizon comparisons. Positive statistics favour the lags + meteorology information set, whereas negative statistics favour lags only. Both panels use the same symmetric colour scale. Open circles mark raw $p < 0.05$ and stars mark station-wise Benjamini–Hochberg retention. No cell survives global Bonferroni correction (0/36); non-marked cells should not be interpreted as evidence of equivalence.

to 22.875 h (+1.000 h), with five of eight lags-only cases already at the boundary and 11 station/condition cases reaching $H^*_{\text{strict}} = 24$ overall. Small changes in H^* in this setting are therefore consistent with limited observable headroom; they do not establish that meteorology contains no incremental information.

The inferential analysis narrows the claim further. Under the primary corrected DM-HLN specification ($q_{\text{overlap}} = 0$), none of the 36 planned comparisons survives global Bonferroni correction, although a small number of within-station Benjamini–Hochberg signals remain. Failure to obtain family-wise rejection is not evidence of equivalence. It instead limits the strength of the multi-site, multi-horizon claim that can be made about attributing the descriptive gains to the added meteorological information.

Dependence in the loss-differential sequence is an important qualification. The choice $q_{\text{overlap}} = 0$ removes only the covariance mechanically required by overlapping multi-step forecast errors; it does not imply independence. The fixed automatic Newey–West sensitivity allows additional positive-lag dependence without selecting the bandwidth from the observed test outcomes. The global result remains 0/36 after Bonferroni correction, while the identity of the limited local signals changes between specifications. This stability at the family-wise level, combined with instability in some local signals, argues against strong claims about station-specific effects.

The evaluation design also has a practical statistical consequence. When forecast origins and physical forecast horizons are expressed on different time scales, the dependence parameters used for DM-HLN/HAC inference must be defined in steps of the loss-differential sequence. Using the physical horizon in hours directly would misrepresent the overlap structure when origins are separated by 24 h.

Several limits determine the scope of these findings. The meteorological covariates are observations available at forecast origin, not future NWP forecasts. The 24-h evaluation range right-censors H^* at several stations, and the empirical evidence is limited to nine sites, the January–July 2023 evaluation window, and the XGBoost design considered here. The automatic Newey–West analysis is a fixed sensitivity rather than an exhaustive model of dependence, and the descriptive comparisons do not support causal attribution. The main implication is therefore to keep three levels of evidence separate: descriptive horizon extension, local inferential signals, and family-wise global support for incremental information.

6 Conclusions

The nine-site results show that descriptive horizon gains and inferential attribution answer different questions. In Madrid, adding meteorological observations extends the strict longest-contiguous positive-skill horizon from 9 to 17 h, while the relaxed measure increases from 15 to 17 h. Across the Irish network, changes are more heterogeneous and are often constrained by the 24-h evaluation boundary.

The 36 planned, dimensionally corrected DM-HLN comparisons provide a more restrictive inferential result. No comparison survives global Bonferroni correction under either the overlap-based specification ($q_{\text{overlap}} = 0$) or the fixed automatic Newey–West sensitivity. That family-wise conclusion is stable across the two dependence specifications, although the identities of the limited within-station BH signals are not.

For multi-horizon environmental forecasting, a longer persistence-relative skill horizon should therefore be interpreted first as a descriptive result. Even a sizeable extension can coexist with limited family-wise evidence that the added information set contributes incremental predictability across the planned site–horizon family.

Table 3 Ireland station-level H^* (hours) by model and criterion. H^*_{strict} is the primary longest-contiguous positive-skill horizon; H^*_{relax} is the secondary last-positive-skill horizon. Entries of 24 for H^*_{strict} indicate reaching the 24-h evaluation boundary ($H_{\max} = 24$).

Station	XGBoost (lags only)		XGBoost (lags + met.)		SARIMA	
	Strict	Relax	Strict	Relax	Strict	Relax
Birr (Co. Offaly)	24	24	24	24	16	24
Dublin Airport	22	24	23	24	24	24
Dundalk (Co. Louth)	24	24	24	24	18	24
Pearse St. Dublin	24	24	24	24	17	23
Ringsend Dublin	24	24	24	24	9	20
Edenderry (Co. Offaly)	16	24	16	24	9	21
Henry St. Limerick	17	24	24	24	4	17
Portlaoise (Co. Laois)	24	24	24	24	17	17
Mean	21.875	24.0	22.875	24.0	14.3	21.3
Mean $\Delta H^*_{\text{strict}}$		+1.000 h			—	

Table 4 Primary DM-HLN inference under the dimensionally coherent overlap-based specification. Forecast origins are spaced by 24 h, so for physical horizons $h \in \{1, 6, 12, 24\}$ the loss-differential horizon in sequence steps is $h_{\text{DM}} = \lceil h/24 \rceil = 1$ and the mechanically induced overlap bandwidth is $q_{\text{overlap}} = \max(0, h_{\text{DM}} - 1) = 0$ for all tests. $q_{\text{overlap}} = 0$ removes mechanically required overlap dependence but does not imply independence. Global Bonferroni is applied across the 36 planned site–horizon tests; Benjamini–Hochberg (BH) is applied within each station (4 horizons) as a local sensitivity analysis.

Specification	Value
Planned comparisons (sites × horizons)	$9 \times 4 = 36$
Valid tests	36/36
Raw $p < 0.05$	3
Global Bonferroni significant	0/36
Station-wise BH significant (count)	3
Station-wise BH identities	Birr ($h = 1$); Madrid ($h = 12$); Pearse St. Dublin ($h = 6$)

Table 5 Fixed automatic Newey–West sensitivity analysis for DM-HLN inference. The HAC bandwidth is set by $q_{\text{auto}} = \lfloor 4\,(n/100)^{2/9} \rfloor$, allowing positive-lag autocovariances in the loss-differential sequence. Global Bonferroni is applied across the 36 planned tests; BH is applied within each station (4 horizons) as a local sensitivity analysis.

Specification	Value
Planned comparisons (sites × horizons)	$9 \times 4 = 36$
Valid tests	36/36
Raw $p < 0.05$	5
Global Bonferroni significant	0/36
Station-wise BH significant (count)	3
Station-wise BH identities	Birr ($h = 1$); Pearse St. Dublin ($h = 6$); Pearse St. Dublin ($h = 12$)

Site	h	Primary $q_{\mathrm{overlap}} = 0$					Automatic Newey–West					
		DM–HLN	p	p_{Bonf}	p_{BH}	BH	q	DM–HLN	p	p_{Bonf}	p_{BH}	BH
Madrid Casa de Campo	1	1.16902	0.243186	1	0.461128	No	5	1.09444	0.274507	1	0.499712	No
Madrid Casa de Campo	6	-0.0501285	0.960048	1	0.960048	No	5	-0.0502606	0.959943	1	0.959943	No
Madrid Casa de Campo	12	2.65199	0.00837127	0.301366	0.0334851	Yes	5	2.45462	0.0145967	0.52548	0.0583866	No
Madrid Casa de Campo	24	0.94394	0.345846	1	0.461128	No	5	0.888673	0.374784	1	0.499712	No
Birr (Co. Offaly)	1	2.68182	0.00790449	0.284561	0.0316179	Yes	4	2.76483	0.00620212	0.223276	0.0248085	Yes
Birr (Co. Offaly)	6	-0.549511	0.583236	1	0.583236	No	4	-0.532491	0.594946	1	0.594946	No
Birr (Co. Offaly)	12	0.889722	0.374628	1	0.583236	No	4	0.805852	0.421235	1	0.594946	No
Birr (Co. Offaly)	24	0.709759	0.478637	1	0.583236	No	4	0.717731	0.473717	1	0.594946	No
Dublin Airport	1	1.6363	0.103961	1	0.415843	No	4	2.06764	0.0404638	1	0.161855	No
Dublin Airport	6	-0.272343	0.785748	1	0.785748	No	4	-0.280021	0.779863	1	0.779863	No
Dublin Airport	12	-0.442456	0.658823	1	0.785748	No	4	-0.435782	0.663648	1	0.779863	No
Dublin Airport	24	0.791337	0.430049	1	0.785748	No	4	1.03687	0.301536	1	0.603072	No
Dundalk (Co. Louth)	1	1.11683	0.265852	1	0.379322	No	4	1.13752	0.257137	1	0.319921	No
Dundalk (Co. Louth)	6	0.916937	0.360638	1	0.379322	No	4	0.997908	0.319921	1	0.319921	No
Dundalk (Co. Louth)	12	0.881718	0.379322	1	0.379322	No	4	1.1321	0.259376	1	0.319921	No
Dundalk (Co. Louth)	24	1.44795	0.149705	1	0.379322	No	4	1.34572	0.180411	1	0.319921	No
Pearse St. Dublin	1	1.84652	0.066217	1	0.0882893	No	4	1.87251	0.0625197	1	0.0833596	No
Pearse St. Dublin	6	2.65868	0.00844603	0.304057	0.0337841	Yes	4	2.94899	0.0035476	0.127714	0.0141904	Yes
Pearse St. Dublin	12	1.8709	0.0627433	1	0.0882893	No	4	2.51309	0.0127155	0.457757	0.025431	Yes
Pearse St. Dublin	24	0.897447	0.370503	1	0.370503	No	4	0.923198	0.356959	1	0.356959	No
Ringsend Dublin	1	1.23326	0.218851	1	0.842041	No	4	1.22665	0.221322	1	0.782339	No
Ringsend Dublin	6	-0.227336	0.820382	1	0.924552	No	4	-0.197207	0.843856	1	0.935797	No
Ringsend Dublin	12	0.806226	0.42102	1	0.842041	No	4	0.859265	0.391169	1	0.782339	No
Ringsend Dublin	24	0.0948146	0.924552	1	0.924552	No	4	0.0806496	0.935797	1	0.935797	No
Edenderry (Co. Offaly)	1	0.339187	0.734813	1	0.775206	No	4	0.335712	0.737428	1	0.806937	No
Edenderry (Co. Offaly)	6	-1.09902	0.273019	1	0.775206	No	4	-1.27485	0.203771	1	0.806937	No
Edenderry (Co. Offaly)	12	0.28594	0.775206	1	0.775206	No	4	0.244688	0.806937	1	0.806937	No
Edenderry (Co. Offaly)	24	0.797665	0.42597	1	0.775206	No	4	0.720269	0.472164	1	0.806937	No
Henry St. Limerick	1	0.917485	0.359941	1	0.479921	No	4	0.817116	0.414788	1	0.553051	No
Henry St. Limerick	6	-1.24178	0.215703	1	0.456771	No	4	-1.40752	0.160753	1	0.462213	No
Henry St. Limerick	12	0.470497	0.638491	1	0.638491	No	4	0.445964	0.656085	1	0.656085	No
Henry St. Limerick	24	1.20805	0.228385	1	0.456771	No	4	1.20099	0.231107	1	0.462213	No
Portlaoise (Co. Laois)	1	1.22102	0.223443	1	0.363908	No	4	1.20507	0.229527	1	0.376582	No
Portlaoise (Co. Laois)	6	-1.0992	0.272931	1	0.363908	No	4	-1.07761	0.282437	1	0.376582	No
Portlaoise (Co. Laois)	12	1.66861	0.0966775	1	0.363908	No	4	1.85433	0.0650878	1	0.260351	No
Portlaoise (Co. Laois)	24	0.679847	0.497347	1	0.497347	No	4	0.703695	0.482398	1	0.482398	No

Table S1 Complete site–horizon DM–HLN results for the fixed paired loss-differential ledger. The sign convention is $d_t = L_{\mathrm{lags}} - L_{\mathrm{lags+meteo}}$, so positive statistics favour the lags + meteorology information set. Primary p-values use $q_{\mathrm{overlap}} = 0$; the sensitivity uses the fixed automatic Bartlett Newey–West bandwidth $q = \lfloor 4(n/100)^{2/9} \rfloor$. Bonferroni correction is across all 36 cells and BH correction is within station. No cell survives global Bonferroni correction.

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Statements and Declarations

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Competing interests

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Author contributions

The author conceived the study, designed the methodology, performed the analyses, developed the software, interpreted the results, and wrote and approved the manuscript.

Data availability

The public GitHub repository https://github.com/fedeg-umh-es/P1_PM10_Meteorology_Hstar contains the versioned analysis code, the processed and regenerated study artifacts archived there, figure-generation support, run metadata, and the final overlap-based and automatic Newey–West inferential tables reported in this manuscript for the underlying meteorology experiment. The Irish station evaluations were regenerated from recovered source datasets using the versioned analysis pipeline. The row-level predictions from the originally executed Irish computational run were not retained; therefore, the reported Irish results correspond to the documented regeneration rather than recovery of the original prediction files. Raw source data are available from the Madrid City Council and the Irish Environmental Protection Agency under their respective open-data terms.

Ethics approval and consent to participate

Not applicable. The study used environmental monitoring data and involved no human participants or animals.

Consent for publication

Not applicable.

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